

Request

Fill the form based on the provided source files ...



Template
Workbook



Source
Files



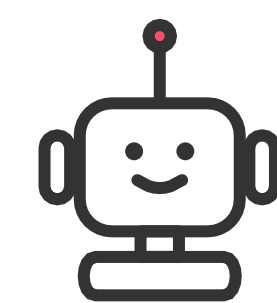
Approval.pdf



Old_card.png



Order_email.eml



Planner

Stepwise Orchestration

Form Filling
Preprocessing

Reflective Region
Adjudication

Need
repair?

Slot Re-grounding and
Local Repair

Verify & Deliver

Need
repair?



Filled

Form Filling Preprocessing

Dual-View Spreadsheet Representation

Executable Spreadsheet Representation



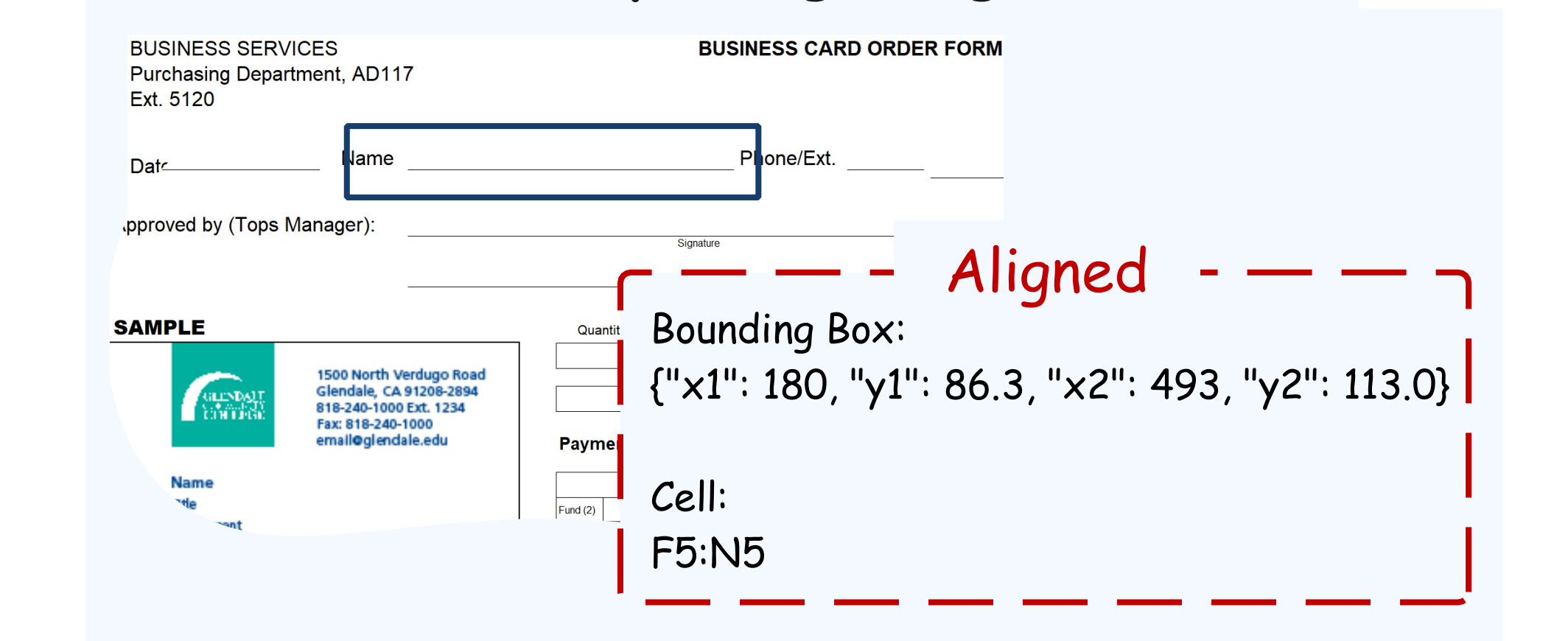
Template
Workbook

Layout Modality Enhancement



Structural
Summary

Corresponding Image



Source File Preprocessing



Source
Files

Type-Based
Content
Extraction

Raw Materials

"headers": {
 "Subject": "Business Card
Order for Robert J. Miller"},
 "body": "Please place a
business card order for Robert J.
Miller."
...}

Structure-Based
Evidence Formatting

- flat key-value pairs
- **sectioned blocks**
- table-oriented records
- hierarchical JSONs
- hybrid section & table
- free-text evidence spans

Statistic Information

content_size	small
kv_line_ratio	0.4167
section_line_ratio	0.0833
bullet_line_ratio	0.0
indent_ratio	0.0
object_depth	2
...	...

Calculate
Structure-Oriented
Information

Source Bundles

"sections": [{ "title":
"root", "content": [{
 "line_index": 1, "text":
"Business Card Order for
Robert J. Miller", ... }] }]
{ "title": "Use the following
contact details", "content":
[{ "line_index": 13, "text":
"Phone: 818-240-1000 Ext.
3420" } ...] }] }

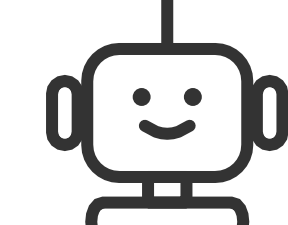
Source Bundle Assembling



Instruction

Fill the form based on the provided source files ...

Prompt

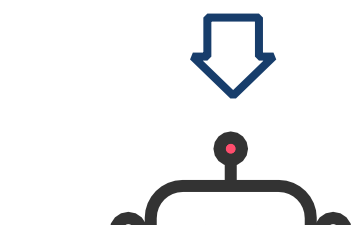


Planner

Prefill Plan

"background_reads": [{
 "stage": "prefill",
 "target": "A8:T11",
 "reason": "Identify approval details from approval.pdf",
 "max_radius": 2 }, ...]

"notes": ["Account number 11-2045-5-601002-4310 needs to be split across Fund, Program, Sub-Program, TOPS, and Object fields.", ...]



Prefill Agent

Prefill

Reflective Region Adjudication

Layout-Grounded
Global Screening



Prefilled



Template

Compare



Filling Actions



Instruction +
Source Bundles

VLM

Coarse Identification

"need_fix": <bool>
"suspicious_cells": ["G10", "K21", ...]
"reliable_cells": ["B5", "G5", ...]

Repair Range Construction

Combined
Identification

Reliable Cells
Suspicious Cells

Expansion

Suspicious Ranges

Reliable
Cells

Suspicious
Ranges

Structure-Aware
Local Adjudication

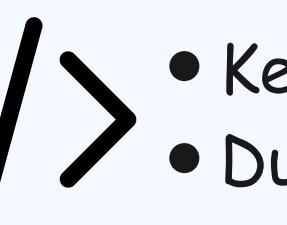


Prefilled



Template

Compare



Representative Risk Signals

- Key/Static-Template Overwrite
- Duplicate Pattern
- Template Inconsistency
- High-Risk Pattern
- Special-Character Anomaly

≥2 Risk Signals?

Yes

Is Suspicious Cell

Quantity
250 for \$18
X 500 for \$28

Payment enclosed or see account number below:
11-2045-5-601002-4310

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

250 for \$18
X 500 for \$28

Payment enclosed or see account number below:
11-2045-5-601002-4310

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

Cells
Pixels

Need Repair?

Yes

Progressive Slot Re-grounding and Repair

Template Restoration

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

Label-Cell Overwrite

Planner

if needed

Account Number

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

Initial Template

Restored

Rectangle-Aware Slot Grounding

Bounding Box: Account Number

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

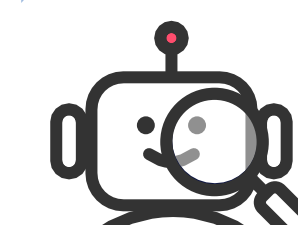
Cell Alignment

Cell-Level Expansion

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)

Cell:

Fund (2)	Program (4)
1: K20	2: L20 3: M20
...	...



RASG

Fillable
Slots
Detection

Local Label-Value Pairs Matching

{ "pairs": [{ "slot_id": "1",
 "label": "Fund (2)",
 "value_cell": "K20", { ... }, ... },
 { "slot_id": "...", { ... }, ... },
 { "slot_id": "...", { ... }, ... },
 { "slot_id": "...", { ... }, ... }] }



VLM

Label-Value Pair Match

Fund (2)	Program (4)
1: K20	

Answer:
K20 corresponds
to Fund (2).

Question: Which label does
cell K20 correspond to?

Summarize

Hints

Prompt

Restored

Planner

Refill Plan

Refill Agent

used or see account number below:
Account Number

Fund (2)	Program (4)	Sub Program (1)	TOPS (6)	Object (4)
11	2045	5	601002	4310

Refilled

Verify & Deliver

Template

↓

Reflective
Region
Adjudication

- Suspicious Ranges
- Reliable Cells

↓

Planner

↓

Latest Suspicious Ranges

↑

Compare

↓

Prior Suspicious Ranges

↓

Need Repair?

Yes

↓

Filled